Hayden Murphey

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SUMMARY

CS graduate with 1.5+ years of contract experience evaluating LLM outputs and debugging AI-generated code: reproducing failures, tracing root causes, and documenting fixes. Hands-on with agent testing, MCP, Langfuse, and Claude Code.

EDUCATION

James Madison University | Bachelor of Science in Computer Science, Minor in Music Graduation: December 2025
Relevant Coursework: Distributed Systems, Operating Systems, Networks & Security, Database Systems, Algorithms, Software Engineering

SKILLS

AI / LLM Systems: Prompt engineering, LLM output evaluation & red-teaming, agentic workflow testing, MCP, LiteLLM, Langfuse, Claude Code, Cursor, Ollama, Whisper
Testing & Debugging: Root-cause analysis, log analysis, REST API testing (Postman), pytest, JUnit, Docker repro environments
Languages: Python, JavaScript/TypeScript, Java, SQL, Bash, C, Rust, Ruby, Haskell, HTML/CSS
Data & Backend: PostgreSQL, SQL Server, Supabase, REST APIs, Node.js, NumPy, pandas
Cloud & DevOps: AWS (EC2, S3, IAM, Lambda, CloudWatch), Proxmox, Ansible, Docker, k3s, GitHub Actions, Git, Linux
Monitoring & Security: Alertmanager, Uptime Kuma, Beszel, ntfy, Caddy, Tailscale, Pi-hole/Unbound, CrowdSec, SOPS/age, TLS
Certifications: AWS Solutions Architect – Associate (Apr 2026) · AWS Cloud Practitioner (Feb 2026) · $(ISC)^2$ Certified in Cybersecurity (Feb 2025) · Machine Learning Specialization, Stanford & DeepLearning.AI (May 2026)

WORK EXPERIENCE

Mercor – AI Output Evaluator, Frontier AI Labs (contract) | Remote May 2026 – Present

- **LLM Output Evaluation & Root-Cause Analysis:** Evaluate AI-generated outputs for frontier AI labs against task requirements, identifying errors and tracing each failure to its root cause.
- **Correction & Documentation:** Review and correct model work, fixing flawed code and responses so they meet spec, and document findings as clear, structured feedback for model training.

DataAnnotation – Software Validation & AI Trainer (contract) | Remote May 2025 – Present

- **Software Quality Engineering:** Validated AI-generated code against technical specifications across 500+ software-engineering and cybersecurity tasks, surfacing logic and edge-case failures that fed into model training.
- **API & Agentic Workflow Testing:** Wrote Python scripts to test REST API integrations and probe agentic-workflow failure modes, improving the reliability of downstream model responses.
- **Issue Reproduction & Regression:** Reproduced bugs in disposable Docker environments and ran regression checks to confirm fixes across languages and frameworks.

Townes Site Engineering – Geotechnical Technician | Richmond, VA May 2021 – August 2025

- **Cross-Functional Communication:** Coordinated daily with field crews, lab staff, and project managers across 40+ projects, relaying technical test results clearly in a fast-paced, client-driven setting.
- **Data Accuracy & Reporting:** Logged 10,000+ test specimens with strict accuracy for on-time client engineering reports.

TECHNICAL PROJECTS

LLM Agent Security Eval Harness (Python, Docker, MCP, LiteLLM, Langfuse) Mar 2026 – Nov 2026

- **Isolated Eval Environment:** Built a Docker benchmark testing whether LLM agents can find and exploit SSRF, with internal-only networks, a host-allowlisted MCP tool, and a fresh random secret per run so the only path to it is the bug.
- **Agent Loop & Tracing:** Wrote a Python agent loop (LiteLLM, 20-step cap) that traces every model and MCP tool call in Langfuse; validated the scorer with a scripted oracle (100%) and a do-nothing baseline (0%).
- **Benchmarking & Failure Analysis:** Ran 50 runs across 2 local models and a 5-level hint ladder, labeling each failure by root cause; no model found the exploit unaided, and Qwen2.5-7B went from 0/5 to 5/5 only when given the exact target.

Homelab Infrastructure as Code (Proxmox, Tailscale, Caddy, Docker) November 2025 – Present

- **Infrastructure & Security:** Proxmox homelab defined as code (4 LXCs, 3 VMs, PBS) with architecture writeup and threat model; no open inbound ports (Tailscale ACLs as code), split-horizon DNS, Caddy wildcard TLS, and CrowdSec.
- **Monitoring & Reliability:** Alerting with Alertmanager, Uptime Kuma, Beszel, and ntfy; DB-consistent backups via vzdump hooks (SQLite, pg_dump) with a retention policy; secrets encrypted in git with SOPS + age.
- **Self-Hosted AI:** Run local LLM inference (Ollama) behind a private API for offline open-weight model experimentation.

Black Box – Language Interpreter & Debugging UI (Ruby) Feb 2025 – May 2025

- **Interpreter & Debugging UI:** Built a Ruby interpreter (lexer, recursive-descent parser, AST evaluator) for a custom language plus a Curses debugging UI with source-indexed error reporting, cutting debug cycles from minutes to seconds.